Yalla Play! — Technical Report

Team 41 · PPNKD · Cycle 3 · Family AI Game Day

1. The problem and our answer

The brief asks for a working website where a family chooses a game, the app hosts it, and an AI judge watches and listens to real play and keeps score fairly. Cycle 3 needs football plus two original games, with both team and individual competition.

Yalla Play! (“Let’s play!”) hosts three games inspired by Emirati life: **Football AI Referee** (team), **Al-Karamah** — setting the Majlis hospitality tray with Dallah, Finjan cups and dates in the right spots and order (individual), and **Dhow Sail Race** — blowing paper dhows down lanes with no hands (team). None of them is one of the four banned examples.

2. User flow (the brief’s six screens)

Step	What the host does	What the app does
1 Choose a game	Picks Football, Al-Karamah or Dhow Race	Opens that game’s own setup
2 Players & teams	Adds P1–P10, names, teams (Team A: P1+P2…), saves	Validates names and teams
3 Rules & ready	Sets points, time, tie-break; ticks everyone ready	Explains points, time limits, penalties, ties and winning in plain words
4 Round in play	Starts 3-2-1-GO, films the play	Shows round, whose turn, challenge, countdown clock and score; plays the GO beep
5 AI judge results	Uploads/records the video, marks goals/lanes/spots, reviews moments	Analyses video + audio, lists each game moment with reason and confidence, verifies the report, applies results
6 Final results	Reads results, picks another game	Winner with reason, match facts, MVP



Home screen: choose a game

3. Architecture

- **Front end:** one self-contained web page (HTML, CSS, vanilla JavaScript, no framework). Runs in any modern browser, on phones and laptops.
- **In-browser AI judge (engine.js):** reads frames from the uploaded video with real seeks, and decodes its sound track with the Web Audio API. Everything runs on the device; the video is never uploaded.
- **Python AI judge (yalla_judge.py):** the same logic in OpenCV + NumPy for long videos, with optional YOLOv8n person/ball detection and ByteTrack tracking.
- **Signed reports:** both judges produce a JSON report signed with HMAC-SHA256 using the match's judge code. The app checks the signature and the video's SHA-256 fingerprint before applying anything.
- **Storage:** finished and in-progress matches are saved in the browser (localStorage), with export to JSON.

4. How the AI judge works

4.1 Vision

- **Colour segmentation:** each frame is converted to HSV. A colour matcher compares hue for colourful clothes and brightness for white/grey/black clothes (kanduras, abayas).
- **Blobs:** matching pixels are grouped with connected-component labelling; shape filters drop pitch lines and noise.
- **Family clothes, not jerseys:** the host clicks each player once. The app samples the more saturated half of the pixels under the click (ignoring printed numbers and logos) to learn that player's clothes colour.
- **Tracking:** players in similar colours share a group; inside a group a slot tracker (one slot per player) matches detections frame to frame with optimal assignment, velocity prediction and a distance gate. Lost-and-found moments are flagged as identity checks.
- **Ball:** small round blob of the ball colour, nearest to its last position.

4.2 Audio

A 1024-point FFT over the sound track finds: **whistles** (strong narrow peak at 2–4.5 kHz), the app's own **countdown beeps** (880 Hz ticks and a 1760 Hz GO tone) and **cheering** (sustained broadband loudness). Because the app plays the GO beep and the camera records it, the AI starts its clock exactly when the family heard GO.



Football AI judge results: tracking boxes, verified report, goal with scorer and assist, possession, player statistics

5. Game logic

Football AI Referee

- **Possession:** the ball belongs to the nearest player if it is close and moving with them (a ball flying past someone is not possession), held for at least two frames.
- **MVP:** the player with the most possession time, as the brief asks. The team with more possession is also shown.
- **Goals:** the ball enters a marked goal box and stays for at least 2 frames. Scorer = last player of the scoring team in possession; assist = the teammate before them. If unclear: “Scorer not identified”.
- **Other moments:** ball out of play, close contact between opponents (flagged for the host, never a foul or card by itself), whistles, cheering.

Al-Karamah

- The host marks a spot on the tray for each item and clicks each item once. The AI measures how much of each spot is filled with each item’s colour, and records when each item is placed and in which spot.
- Points: +10 per item in place, +10 for the right serving order, +5 inside the time limit, minus penalties for wrong spot or order (all editable).

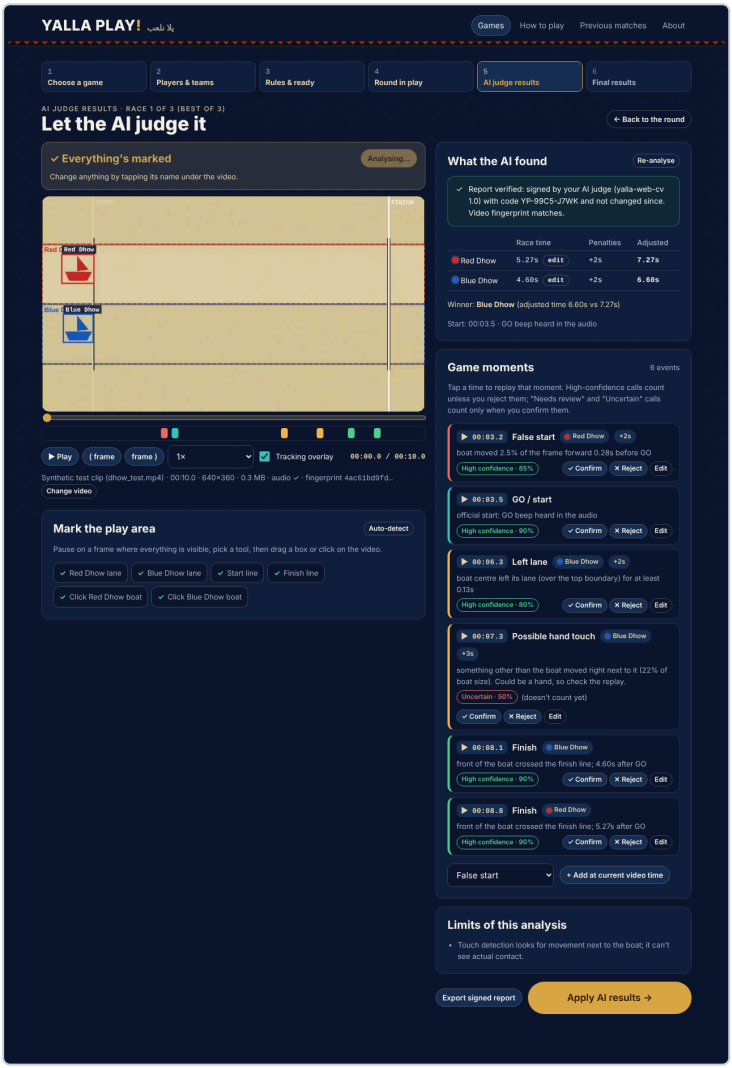
Dhow Sail Race

- Each boat is tracked by colour inside its lane. Race time = front of the boat crossing the finish line minus the GO beep time, interpolated between frames.

- Penalties: false start (moved before GO) +2 s, leaving the lane +2 s, possible hand touch (motion next to the boat that isn't the boat) +3 s. Adjusted time = race time + confirmed penalties; best of 3.

6. Fairness and the host in charge

- Every AI moment has a timestamp, a written reason and a confidence label: High confidence, Needs review or Uncertain.
- High-confidence calls count unless the host rejects them; everything else counts only when the host confirms it.
- Tap any moment to replay it (2.5 s before to 2 s after) with slow motion, frame stepping and tracking boxes.
- The host can confirm, reject, edit (team, player, type, time, penalty) or add missed moments; corrections are recorded separately from the AI's signed facts.
- No face recognition: players are known only by clicks, clothes colours and labels.



Dhow Race judge: false start, lane exit, possible touch (Uncertain, not counted until confirmed), adjusted times

7. Two ways to connect the AI judge

In the app: press ANALYSE. A progress list shows real stages (for football: preparing video and audio, detecting players and the ball, matching players to their clothes, tracking, detecting goals and rule moments, creating timestamps, generating the report) with genuine percentages.

Python: download `yalla_config.json` (the markings) and run `python yalla_judge.py football match.mp4 --config yalla_config.json --key YP-XXXX-XXXX`. Import the signed report into the app.

Integrity check: the payload is signed with HMAC-SHA256 keyed by the judge code. Changing a single character (we tested changing a goal's team from A to B) makes the app reject it: "Signature

doesn’t match”. The report also carries the video’s SHA-256, so a report about a different video is refused.



Python judge report imported and verified in the app

8. Testing

Test	Expected	Result
Football synthetic clip (mixed clothes: red, purple, white, blue)	Goal at 0:06.6 by P2, assist P1; whistle at 0:07	Pass (browser and Python)
Dhow synthetic clip	False start, lane exit, touch, finish ≈ 5.3 s and 4.6 s	Pass (5.27 s / 4.60 s)
Al-Karamah synthetic clip	Dallah → Dates → Cups (wrong order), complete in 6.5 s	Pass
Tampered report	Rejected	Pass
Phone width (400 px)	No sideways scrolling	Pass
Add player / open page with no server	Works	Pass after fixes

All flows were run automatically in a headless browser (Playwright). Test clips are clearly labelled as computer-generated; the demo video uses real family play.

9. Limitations

- Detection is by colour: a green outfit on grass, or a ball the same colour as someone’s clothes, confuses it. We recommend a bright orange ball.
- Players in matching clothes are told apart by movement; when they cross, identity can swap (the app flags these moments).
- One camera can’t judge offside or handball, and touch detection sees motion, not actual contact. These are left to the host.
- Camera recording is blocked inside the Claude page view; it works when the app is run locally. Uploading a phone video always works.
- The optional YOLO mode was written but not tested in our build environment.

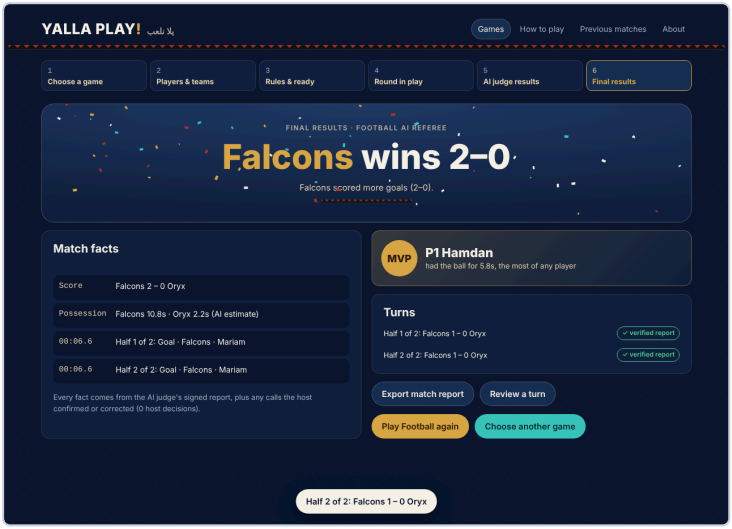
10. What we learned

- **Honest AI beats impressive AI.** Showing reasons and confidence, and letting the host confirm, made the judge feel fair.
- **Design for real families.** Our first version assumed jerseys; family games are played in kanduras, abayas and t-shirts, so we changed to per-player clothes colours.
- **Sound is a free sensor.** Making the app play a known GO tone gave the AI an exact start time with simple FFT analysis.

- **Trust needs proof.** Signing reports with HMAC showed us how real systems prevent tampering.
- **Test like a user.** Our own testing found three bugs (clothes, Add player scrolling, test clip loading) that automated tests had missed.

11. Setup and run

- Open the prototype link, or open `index.html` in Chrome (works with no install).
- For camera recording: in the app folder run `python -m http.server 8000` and open `localhost:8000`.
- Python judge: `pip install opencv-python numpy` (+ ffmpeg for audio, optional `ultralytics`).



Final results: winner with reason, match facts, MVP